

31338 Network Servers

Lab 4a: Configuring a DHCP client

Aims

1. To configure a Linux machine as a DHCP client
2. To configure a Windows machine as a DHCP client

Background Information

VMware is setup to provide a DHCP service to client VMs when enabled. Our configuration has a virtual DHCP server for network VMnet8, the default NAT interface. This DHCP server will assign IP addresses in the range 192.168.3.128 – 192.168.3.254 inclusive if you are in the lab room (it may be different if you are at home).

But for other machines and other interfaces (like our VMnet2 network), you may need to know how to configure DHCP client workstations. We will also need this in later Lab 4 activities.

Also note that a lot of internet documentation about Linux DHCP refers to a separate 'dhclient' program. When using NetworkManager (as we are), there is no need to run dhclient separately. NetworkManager has DHCP support built in.

Task 1: Linux command-line configuration

Test the network connectivity using `ping` and `ssh`. Check your machine's IP address with `ifconfig` or `nmcli`.

Verify and/or change the configuration files so that when your machine boots, it will automatically send a DHCP query to retrieve network settings. The most important change is to change `BOOTPROTO` from "none" to "dhcp" in the interface-specific configuration file in `/etc/sysconfig/network-scripts/ifcfg-ens*` (where `ens*` is your specific adapter eg: `ens37`).

But in addition to that change, also comment out the other lines that manually set the IP address, netmask, etc. in both the `ifcfg-ens*` file, and the line that sets the interface as the default gateway. The comment character is a `#` (same as a UNIX shell script).

Reboot your virtual machine and test that the network is up and that the IP address appears to have been dynamically allocated. Test network connectivity using `ping` and `ssh`. Alternatively you could just `ifdown` and `ifup`, but rebooting ensures all caches, etc are cleared. But on a production server you wouldn't normally reboot for testing!

Note: from previous lab exercises where you installed software updates, you might have already configured your machine as a DHCP client without even knowing it. So if the above description seems like it already matches what is on your system, just verify it and make sure you understand what's happening (and make notes in your journal).

Task 2: Linux GUI configuration

You can also configure this via the Settings icon → Network menu item, or running either `gnome-control-center` or `nm-connection-editor`

Choose the appropriate adapter, and then Click the properties to change this. Choose the appropriate tab to change the configuration (usually IPv4 settings for DHCP).

HINT: If you haven't already done so, you can also take this time to make `ens37` connect automatically so you don't have to keep clicking the network manager icon to start the network connection!

Task 3: Windows configuration

On Windows Server, you use the GUI to do most configuration, although it is useful to know the command-line versions to create scripts to automate management of servers (or client workstations).

1. From Server Manager, choose Server Manager → View Network Connections (alternative: Start → Admin Tools → Settings → Control Panel → Network and Sharing centre → Manage Network Connections)
2. Choose the appropriate adapter & select Properties
3. In the Items listing, choose Internet Protocol v4 (TCP/IP) and choose Properties
4. Set the “Obtain an IP address automatically” radio button

You can also investigate how to do this from the command line. You need to use the following command
`netsh`

Some helpful subcommands:

Show configuration:

```
netsh interface ip show config
```

Set interface to use DHCP:

```
netsh interface ip set address "Local Area Connection" dhcp
```

Set interface to use static IP address:

```
netsh interface ip set address name="Local Area Connection" static 10.0.2.11 255.255.255.0 10.0.2.1
```

would set Local Area Connection to 10.0.2.11 with mask 255.255.255.0 and default gateway 10.0.2.1

See: <http://support.microsoft.com/kb/257748>

Explore other settings of netsh, a quick way to view everything is to use:

```
netsh dump
```

Please note the most useful commands into your learning journal. You may need these when you do the skills test!